26900102 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCHE300 - ORGANIC CHEMISTRY - 3

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which reagent selectively reduces only one nitro group in polynitro compounds? [K] / [CO1]

   a). Sn/HCl

   b). NaOH/Glucose

   c). Raney Ni

   d). NaSH

2. What is the hybridization of the carbon atom in the aci-nitro form of nitromethane? [U] / [CO1]

   a). sp

   b). sp^2

   c). sp^3

   d). None of the above

3. An alkyl halide reacts with KCN and AgCN separately. The products are respectively: [U] / [CO1]

a). Isocyanide and nitrile b). Nitrile and isocyanide c). Amide and nitrile d). Nitrile and amide

4. The product obtained by reduction of CH_3CN with Stannous chloride /HCl followed by steam distillation is [U] / [CO1]

a). CH_3NH_2 b). HCHO c). CH3CHO d). CH_4

5. Benzenediazonium chloride reacts with CuBr to give: [U] / [CO1]

a). Bromobenzene b). Chlorobenzene c). Fluorobenzene d). Phenol

6. Nitration of benzoic acid predominantly gives: [U] / [CO1]

a). *o*-Nitrobenzoic acid

b). *p*-Nitrobenzoic acid

c). *m*-Nitrobenzoic acid

d). 2,4-Dinitrobenzoic acid

7. The reaction between sodium ethoxide and methyl iodide gives: [U] / [CO2]

a). Methanol b). Diethyl ether

c). Dimethyl ether d). Ethyl methyl ether

8. [U] / [CO3]
The pyrimidine base characteristic of RNA is:

a). Adenine b). Thymine c). Uracil d). Guanine

9. [U] / [CO5]
Rhodopsin consists mainly of:

a). Retinal and opsin b). Retinol and glucose c). Vitamin C and opsin d). Carotene and protein

10. [U] / [CO5]
Paternò–Büchi reaction involves the reaction of a carbonyl compound with:

a). Alkane b). Alkene c). Alkyne only d). Aromatic hydrocarbon

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Identify the major products and write the chemical equations for the reduction of nitrobenzene using: [A] / [CO1]
    a) Zn dust/NH_4Cl
    b) Zn dust/NaOH

12. Compare Gabriel's phthalimide synthesis and the Hoffmann bromamide reaction with respect to starting materials, reagents, mechanism, and products. [U] / [CO1]

13. A compound has a five-membered ring containing one oxygen atom and two C=C bonds. Identify the compound and draw its structure. Explain why it is classified as an aromatic heterocycle. [A] / [CO3]

14. Discuss the mechanism of ketonic hydrolysis of ethyl acetoacetate [U] / [CO4]

15. Define electronic excitation and explain its significance in photochemistry. [U] / [CO5]

16. What is a Norrish Type I reaction? Explain with a suitable example. [U] / [CO5]

[5x4 = 20]

Part C

Short Essay

Answer any **2** questions

Each question carries **10** marks

17. Explain the complete procedure for identifying an unknown amine using the Hinsberg test. Include the observations and chemical reactions involved. [U] / [CO1]

18. Write a brief note on the structure of epoxides and the various reactions of epoxides [U] / [CO2]

19. a) Give one method of preparation of each of: i) Furan, iii) Pyrrole and iii) Thiophene. [An] / [CO3]
    b) What happens when pyridine reacts with sodamide? Name the reaction.

20. a) Explain the reaction of diethyl malonate with sodium ethoxide followed by an alkyl halide. Discuss the mechanism. [U] / [CO4]
    b) Describe the synthesis of acetic acid using diethyl malonate.

[10x2 = 20]